Linear Algebra 2

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Abstract

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6 Lecture 6

How to relate different vector spaces.

6.1 Linear maps

Definition 6.1

Linear map

Let V and W be vector spaces over a field $\mathbb{F}$. A map $T : V \to W$ is linear if for all $u, v \in V$ and $\alpha \in \mathbb{F}$,

$$T(u+v) = T(u) + T(v), \quad T(\alpha v) = \alpha T(v).$$

Additivity and homogeneity

- Additivity: $T(u+v) = T(u) + T(v)$ for all $u, v \in V$.
- Homogeneity: $T(\alpha v) = \alpha T(v)$ for all $\alpha \in \mathbb{F}, v \in V$.

Notation

- $\mathcal{L}(V, W) =$ the space of linear maps $T : V \to W$
- $\mathcal{L}(V) =$ the space of linear maps $T : V \to V$

6.2 Examples

6.2.1 The zero linear map

$$\begin{aligned} 0 \colon V &\to W, \\ v &\mapsto 0. \end{aligned}$$

Remark

$T(0_V) = 0_W$.

$$\begin{aligned} 0_V &= 0_{\mathbb{F}} v, \\ T(0_V) &= T(0_{\mathbb{F}} v) = 0_{\mathbb{F}} T(v) = 0_W. \end{aligned}$$

6.2.2 The identity linear map

For any $v \in V$,

$$\begin{aligned} I \colon V &\to V, \\ v &\mapsto v. \end{aligned}$$

6.2.3 Differentiation operator

First recall the definition of a polynomial space over $\mathbb{R}$.

$$\mathcal{P}(\mathbb{R}) = \{\text{polynomials } a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \mid a_i \in \mathbb{R}, n \in \mathbb{N}\}$$

Then define the differentiation operator.

$$\begin{aligned} D \colon \mathcal{P}(\mathbb{R}) &\to \mathcal{P}(\mathbb{R}) \\ \psi &\mapsto \psi' \end{aligned}$$

For $\psi \in \mathcal{P}(\mathbb{R})$,

$$\begin{aligned} \psi &= a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n, \\ \psi' &= a_1 + 2a_2 x + \cdots + n a_n x^{n-1}. \end{aligned}$$

For all $\alpha \in \mathbb{R}$ and all polynomials $p, q \in \mathcal{P}(\mathbb{R})$,

$$\begin{aligned} (p + q)' &= p' + q', \\ (\alpha p)' &= \alpha p'. \end{aligned}$$

The differentiation operator is linear.

6.2.4 Integration operator

Define the integration operator.

$$\begin{aligned} \int : \mathcal{P}(\mathbb{R}) &\to \mathcal{P}(\mathbb{R}) \\ \psi &\mapsto \int \psi \, dx \end{aligned}$$

For all $\alpha \in \mathbb{R}$ and all polynomials $p, q \in \mathcal{P}(\mathbb{R})$,

$$\begin{aligned} \int (p+q) &= \int p + \int q, \\ \int (\alpha p) &= \alpha \int p. \end{aligned}$$

6.2.5 Multiplication by a fixed polynomial

Define the multiplication by a fixed polynomial operator.

$$\begin{aligned} T : \mathcal{P}(\mathbb{R}) &\to \mathcal{P}(\mathbb{R}) \\ \psi &\mapsto q\psi \end{aligned}$$

So

$$(Tp)(x) = q(x)p(x).$$

Check that T is linear.

6.2.6 Backward shift operator

Define the backward shift operator.

$$\begin{aligned} T : \mathbb{F}^\infty &\to \mathbb{F}^\infty \\ (x_1, x_2, \dots) &\mapsto (x_2, x_3, \dots) \end{aligned}$$

Check that T is linear.

6.2.7 Forward shift operator

Define the forward shift operator.

$$\begin{aligned} T : \mathbb{F}^\infty &\to \mathbb{F}^\infty \\ (x_1, x_2, \dots) &\mapsto (0, x_1, x_2, \dots) \end{aligned}$$

Check that T is linear.

6.2.8 Linear map and matrices

$$\begin{aligned} &T : \mathbb{F}^n \to \mathbb{F}^m, \\ &\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \mapsto \begin{bmatrix} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n \\ \vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n \end{bmatrix}. \end{aligned}$$

T should be a matrix.

$$\begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix}_{(m\times n)} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$m \times n$ matrix: rows $\times$ columns.

$$T\begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \; T\begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \; \dots, \; T\begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Columns!
So, because the lemma below shows that a linear map is completely determined by its action on the basis vectors of the domain. Any linear map can be uniquely represented by a matrix with respect to the chosen bases.

6.3 A linear map is determined by its values on a basis

Lemma 6.1

A linear map $T\colon V \to W$ is uniquely determined by what it does on a basis of V.

Example

For a basis $\{v_1, \dots, v_n\}$ of V, let

$$T(v_i) = w_i \in W \quad \text{for } i = 1, 2, \dots, n.$$

$$\begin{aligned} T(v_1) &= w_1, \\ T(v_2) &= w_2, \\ &\vdots \\ T(v_n) &= w_n. \end{aligned}$$

Then T is uniquely determined by the images of the basis vectors.

$$\begin{bmatrix} w_1 & w_2 & \dots & w_n \end{bmatrix}$$

determines T uniquely. For $v \in V$,

$$\begin{aligned} Tv &= T\left(a_1v_1 + a_2v_2 + \cdots + a_nv_n\right) \\ &= a_1T(v_1) + a_2T(v_2) + \cdots + a_nT(v_n) \\ &= a_1w_1 + a_2w_2 + \cdots + a_nw_n. \end{aligned}$$

Check that T is a linear map.

6.3.1 $\mathcal{L}(V, W)$

Let $S, T \in \mathcal{L}(V, W)$.

$$\begin{aligned} (S+T)(v) &= S(v) + T(v) \quad && \text{for all } v \in V, \\ (\lambda S)(v) &= \lambda S(v) && \text{for all } v \in V \text{ and } \lambda \in \mathbb{F}. \end{aligned}$$

Claim

(Need to check)

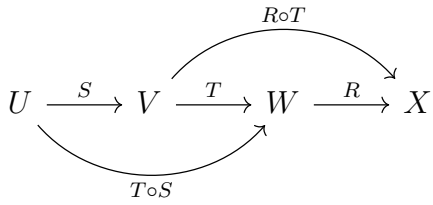
$$S + T, \lambda S \quad \text{are linear maps.}$$

That makes $\mathcal{L}(V, W)$ a vector space over $\mathbb{F}$.
Similarly, $\mathcal{L}(V)$ is a vector space over $\mathbb{F}$ for any vector space V.

6.3.2 Compositions of linear maps

Case 1

Let U, V, W be vector spaces over $\mathbb{F}$.



For all $u \in U$,

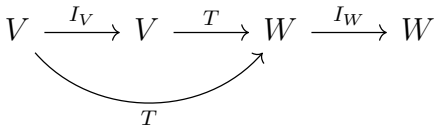
$$\begin{aligned} (T \circ S)(u) &= T(S(u)), \\ (R \circ (T \circ S))(u) &= R(T(S(u))), \\ ((R \circ T) \circ S)(u) &= (R \circ T)(S(u)) = R(T(S(u))). \end{aligned}$$

Therefore,

$$R \circ (T \circ S) = (R \circ T) \circ S.$$

Case 2

For $T \colon V \to W$.



So

$$\begin{aligned} T \circ I_V &= T, \\ I_W \circ T &= T. \end{aligned}$$

For $S, T \colon V \to V$, the compositions $S \circ T$ and $T \circ S$ need not agree.

Example

For $V = \mathcal{P}(\mathbb{R})$,

$$\begin{aligned} T\colon \mathcal{P}(\mathbb{R}) &\to \mathcal{P}(\mathbb{R}) \\ p(x) &\mapsto p'(x) \end{aligned}$$

$$\begin{aligned} S\colon \mathcal{P}(\mathbb{R}) &\to \mathcal{P}(\mathbb{R}) \\ p(x) &\mapsto xp(x) \end{aligned}$$

$T(p(x)) = p'(x), S(p(x)) = xp(x)$.

$$\begin{aligned} (S \circ T)(p(x)) &= S(p'(x)) = xp'(x), \\ (T \circ S)(p(x)) &= T(xp(x)) = p(x) + xp'(x). \end{aligned}$$

6.4 Kernel and Image of a linear map

Definition 6.2

Kernel and Image Spaces

Let $T\colon V \to W$ be a linear map.

$$\begin{aligned} \ker T &= \{v \in V \mid T(v) = 0_W\} && \text{kernel of } T \text{ (null space of } T), \\ \operatorname{Im} T &= \{T(v) \mid v \in V\} && \text{image of } T \text{ (range of } T). \end{aligned}$$

Example

Zero map

For the zero map,

$$\begin{aligned} 0\colon V &\to W \\ v &\mapsto 0 \end{aligned}$$

We can immediately see that the range of the zero map is just the zero vector in W, and the null space is the entire space V.

$$\begin{aligned} \operatorname{Im}(0) &= \{0\}, \\ \ker(0) &= V. \end{aligned}$$

Example

Differentiation map

For the differentiation map,

$$D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$$
$$\gamma \mapsto \gamma'$$

We can immediately see that the range of the differentiation map is the set of all derivatives of functions in $\mathcal{P}(\mathbb{R})$, and the null space is the set of all constant polynomials.

$$\operatorname{Im}(D) = \{\gamma' \mid \gamma \in \mathcal{P}(\mathbb{R})\},$$
$$\ker(D) = \{\text{constant polynomials}\}.$$

7 Lecture 7

7.1 Properties of Linear Maps

Let V and W be vector spaces over the same field $\mathbb{F}$. Let $T\colon V \to W$ be a linear map. For all $v_1, v_2 \in V$ and $\alpha \in \mathbb{F}$,

$$\begin{cases} T(v_1 + v_2) = T(v_1) + T(v_2), \\ T(\alpha v_1) = \alpha T(v_1), \end{cases} \iff T(u + \alpha v) = T(u) + T(\alpha v) = T(u) + \alpha T(v).$$

7.2 Injective and Surjective Maps

7.2.1 Kernel and Image Definition

Definition 7.1

The kernel of T

Let $T\colon V \to W$ be a linear map.

$$\ker T = \{v \in V \mid T(v) = 0_W\}.$$

Definition 7.2

The image of T

Let $T\colon V \to W$ be a linear map.

$$\operatorname{Im} T = \{T(v) \mid v \in V\}.$$

Theorem 7.1

The kernel and image of a linear map T are vector subspaces of V and W, respectively.

$$\ker T \subseteq V,$$
$$\operatorname{Im} T \subseteq W.$$

Proof. Since $T(0_V) = 0_W$, the kernel contains 0_V.
If $v_1, v_2 \in \ker T$ and $\lambda \in \mathbb{F}$, then

$$T(v_1 + \lambda v_2) = T(v_1) + \lambda T(v_2) = 0_W,$$

so $v_1 + \lambda v_2 \in \ker T$.
Thus $\ker T$ is a subspace of V. □

Proof. If $w_1, w_2 \in \operatorname{Im} T$, write

$$w_1 = T(v_1),$$
$$w_2 = T(v_2)$$

for some $v_1, v_2 \in V$. For $\lambda \in \mathbb{F}$,

$$w_1 + \lambda w_2 = T(v_1) + \lambda T(v_2) = T(v_1 + \lambda v_2) \in \operatorname{Im} T.$$

Thus $\operatorname{Im} T$ is a subspace of W. □

7.2.2 Injective

Definition 7.3

The linear map $T\colon V \to W$ is called *injective* (or one-to-one) if $\ker T = \{0_V\}$.

For $T\colon V \to W$, injectivity is equivalent to

$$T(v_1) = T(v_2) \implies v_1 = v_2.$$

Indeed, if $T(v_1) = T(v_2)$, then

$$T(v_1 - v_2) = 0_W,$$

so

$$v_1 - v_2 \in \ker T = \{0_V\},$$

and hence

$$v_1 = v_2.$$

Conversely, if T is injective and $v \in \ker T$, then

$$T(v) = T(0_V),$$

so $v = 0_V$.

7.2.3 Surjective

Definition 7.4

The linear map $T\colon V \to W$ is called *surjective* (or onto) if $\operatorname{Im} T = W$.

If T is surjective, then by definition $\operatorname{Im} T = W$.
Equivalently, for every $w \in W$ there exists $v \in V$ such that

$$w = T(v).$$

Example

The derivative map

$$D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$$
$$p \mapsto p'$$

has

$$\ker D = \{p \in \mathcal{P}(\mathbb{R}) \mid D(p) = 0\},$$
$$\operatorname{Im} D = \{D(p) \mid p \in \mathcal{P}(\mathbb{R})\}.$$

D is surjective but not injective.

Example

The backward shift b and forward shift f on $\mathbb{F}^\infty$ are defined by

$$\mathbb{F}^\infty \underset{f}{\overset{b}{\rightleftarrows}} \mathbb{F}^\infty$$

$$b(x_1, x_2, x_3, \dots) = (x_2, x_3, x_4, \dots),$$
$$f(x_1, x_2, x_3, \dots) = (0, x_1, x_2, \dots).$$

The kernels and images of the backward and forward shifts are given by

$$\ker b = \{(x_1, x_2, x_3, \dots) \in \mathbb{F}^\infty \mid x_2 = x_3 = \cdots = 0\},$$
$$\operatorname{Im} b = \mathbb{F}^\infty,$$
$$\ker f = \{0\},$$
$$\operatorname{Im} f = \{(x_1, x_2, x_3, \dots) \in \mathbb{F}^\infty \mid x_1 = 0\}.$$

The backward shift is surjective but not injective, while the forward shift is injective but not surjective.

Example

$$T\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R}),$$
$$p(x) \mapsto xp(x).$$
$$\ker T = \{p(x) \in \mathcal{P}(\mathbb{R}) \mid xp(x) = 0\} = \{0\}.$$
$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n, \quad a_i \in \mathbb{R}.$$
$$xp(x) = a_0 x + a_1 x^2 + a_2 x^3 + \cdots + a_n x^{n+1}.$$
$$\operatorname{Im} T = \{xp(x) \mid p(x) \in \mathcal{P}(\mathbb{R})\} \neq \mathcal{P}(\mathbb{R}).$$

T is injective but not surjective.

7.3 The Fundamental Theorem of Linear maps

Theorem 7.2

The fundamental theorem of linear maps (also known as the rank-nullity or rank theorem)

If $T\colon V \to W$ is a linear map, and the dimension of V is finite, then

$$\dim V = \underbrace{\dim \ker T}_{\text{nullity}} + \underbrace{\dim \operatorname{Im} T}_{\text{rank}}.$$

Proof. Since $\ker T \subseteq V$, let $\{u_1, \dots, u_n\}$ be a basis for $\ker T$. Extend it to a basis for V.

$$\{u_1, \dots, u_n, v_1, \dots, v_r\}$$

Thus

$$\dim V = n + r, \quad \dim \ker T = n.$$

We want to show that $\dim \operatorname{Im} T = r$, so we claim that $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\operatorname{Im} T$, need to be proven.

Claim: $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\operatorname{Im} T$.

1. $\{T(v_1), \dots, T(v_r)\}$ spans $\operatorname{Im} T$. Since $\{u_1, \dots, u_n, v_1, \dots, v_r\}$ is a basis for V, any $v \in V$ can be written as

   $$v = a_1u_1 + \cdots + a_nu_n + b_1v_1 + \cdots + b_rv_r.$$

   Applying T, we get

   $$T(v) = b_1T(v_1) + \cdots + b_rT(v_r),$$

   since $T(u_i) = 0$ for all i. Therefore, $\{T(v_1), \dots, T(v_r)\}$ spans $\operatorname{Im} T$.

2. $\{T(v_1), \dots, T(v_r)\}$ is linearly independent. Suppose

   $$c_1T(v_1) + \cdots + c_rT(v_r) = 0.$$

   Then

   $$T(c_1v_1 + \cdots + c_rv_r) = 0.$$

   Therefore, $c_1v_1 + \cdots + c_rv_r \in \ker T$. But $\{u_1, \dots, u_n, v_1, \dots, v_r\}$ is a basis for V, so $c_1v_1 + \cdots + c_rv_r$ can be written uniquely as a linear combination of the basis vectors:

   $$c_1v_1 + \cdots + c_rv_r = a_1u_1 + \cdots + a_nu_n.$$

   Hence

   $$c_1v_1 + \cdots + c_rv_r - (a_1u_1 + \cdots + a_nu_n) = 0.$$

   Since the basis vectors are linearly independent, we must have

   $$c_1 = \cdots = c_r = 0.$$

   This proves that $\{T(v_1), \dots, T(v_r)\}$ is linearly independent, completing the proof of the claim. Therefore, $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\operatorname{Im} T$.

$$\dim \operatorname{Im} T = r, \quad \dim \ker T = n, \quad \dim V = n + r.$$

$$a_1 = \cdots = a_n = 0, \quad c_1 = \cdots = c_r = 0.$$

This proves that $\{T(v_1), \ldots, T(v_r)\}$ is linearly independent, completing the proof of the claim.

Therefore, $\{T(v_1), \ldots, T(v_r)\}$ is a basis for $\operatorname{Im} T$.

$$\dim \operatorname{Im} T = r, \quad \dim \ker T = n, \quad \dim V = n + r.$$

□

7.4 Corollaries of the Rank Theorem

Let $T\colon V \to W$ with

1. $\dim V < \infty$
2. $\dim W < \infty$.

7.4.1 Injective Maps

If T is injective, then

$$\dim V = \dim \operatorname{Im} T \leq \dim W,$$

because $\ker T = \{0_V\}$.
So if $\dim V > \dim W$, no injective map $T\colon V \to W$ exists.

7.4.2 Surjective Maps

If T is surjective, then

$$\dim W = \dim \operatorname{Im} T \leq \dim V,$$

because $\operatorname{Im} T = W$.
So if $\dim V < \dim W$, no surjective map $T\colon V \to W$ exists.

7.4.3 Bijective Maps

If $T\colon V \to W$ is injective and surjective, then $\dim V = \dim W$.

Proof. Since T is injective,

$$\dim V = \dim \operatorname{Im} T \leq \dim W.$$

Since T is surjective,

$$\dim W = \dim \operatorname{Im} T \leq \dim V.$$

Therefore,

$$\dim V = \dim W.$$

□

8 Lecture 8

8.1 Review

For $T: V \to W$,

$$\ker T = \{v \in V \mid T(v) = 0_W\} \subseteq V,$$
$$\operatorname{Im} T = \{T(v) \mid v \in V\} \subseteq W.$$

- T is injective if $\ker T = \{0_V\}$.
- T is surjective if $\operatorname{Im} T = W$.

8.2 Invertible Maps

Definition 8.1

A linear map $T: V \to W$ is *invertible* if there exists a linear map $S: W \to V$ such that

$$S \circ T = I_V \quad \text{and} \quad T \circ S = I_W.$$

S is an *inverse* of T.

Theorem 8.1

An invertible linear map $T: V \to W$ has a unique inverse.

Proof. Suppose T has two inverses S_1 and S_2. Then

$$T \circ S_1 = I_W \quad \text{and} \quad T \circ S_2 = I_W,$$

$$S_1 \circ T = I_V \quad \text{and} \quad S_2 \circ T = I_V.$$

Therefore,

$$S_1 = S_1 \circ I_W = S_1 \circ (T \circ S_2) = (S_1 \circ T) \circ S_2 = I_V \circ S_2 = S_2.$$

Hence $S_1 = S_2$, proving the uniqueness of the inverse. □

Theorem 8.2

If $T: V \to W$ is a linear map, then

$$T \text{ is invertible} \iff T \text{ is both injective and surjective.}$$

Proof. Proof in both directions.

1. ($\Rightarrow$) If T is invertible, then it is both injective and surjective.
   Show T is injective
   Let $v \in V$ such that $T(v) = 0_W$, then

   $$v = T^{-1}T(v) = T^{-1}(0_W) = 0_V.$$

   Thus

   $$\ker T = \{0_V\} \implies T \text{ is injective.}$$

Show T is surjective
Let $w \in W$, then

$$w = T(T^{-1}(w)) \implies w \in \operatorname{im} T.$$

So

$$W \subseteq \operatorname{im} T \implies \operatorname{im} T = W \implies T \text{ is surjective.}$$

2. ($\Leftarrow$) Conversely, if T is both injective and surjective, then it is invertible.
Since T is surjective, $\forall w \in W$, $\exists v \in V$ such that $T(v) = w$.
Define

$$T^{-1} \colon W \to V,$$
$$w \mapsto v$$

This is well-defined because T is injective.
Then

$$T^{-1} \circ T(v) = v \quad \text{and} \quad T \circ T^{-1}(w) = T(v) = w$$

so

$$T^{-1} \circ T = I_V \quad \text{and} \quad T \circ T^{-1} = I_W$$

so T is invertible.

□

Claim T^{-1} is linear. Let $w_1, w_2 \in W$ and $\alpha, \beta \in \mathbb{F}$. Then

$$\begin{aligned} T^{-1}(\alpha w_1 + \beta w_2) &= T^{-1}(\alpha T(v_1) + \beta T(v_2)) \\ &= T^{-1}(T(\alpha v_1 + \beta v_2)) \\ &= \alpha v_1 + \beta v_2 \\ &= \alpha T^{-1}(w_1) + \beta T^{-1}(w_2) \end{aligned}$$

where $v_1 = T^{-1}(w_1)$ and $v_2 = T^{-1}(w_2)$. Hence T^{-1} is linear.

Example

Consider the forward and backward linear maps

$$b \colon \mathbb{F}^\infty \to \mathbb{F}^\infty,$$
$$(x_1, x_2, \dots) \mapsto (x_2, x_3, \dots)$$

$$f \colon \mathbb{F}^\infty \to \mathbb{F}^\infty,$$
$$(x_1, x_2, \dots) \mapsto (0, x_1, x_2, \dots)$$

b is surjective but not injective, f is injective but not surjective $\implies$ neither b nor f is invertible.

$$b \circ f = I_{\mathbb{F}^\infty} \quad \text{but} \quad f \circ b \neq I_{\mathbb{F}^\infty}.$$

This cannot happen for finite-dimensional vector spaces.

Theorem 8.3

Let $T \colon V \to W$ be a linear map with $\dim V = \dim W < \infty$. Then

$$T \text{ is injective} \iff T \text{ is surjective} \iff T \text{ is invertible.}$$

Proof. By the rank-nullity theorem, we have

$$\dim V = \dim \ker T + \dim \operatorname{Im} T = \dim W.$$

$$\begin{aligned} T \text{ injective} &\implies \dim \ker T = 0 \\ &\implies \dim \operatorname{Im} T = \dim V = \dim W \\ &\implies T \text{ is surjective.} \end{aligned}$$

$$\begin{aligned} T \text{ surjective} &\implies \dim \operatorname{Im} T = \dim W = \dim V \\ &\implies \dim \ker T = 0 \\ &\implies T \text{ is injective.} \end{aligned}$$

Hence T is invertible. □

Theorem 8.4

Let $T\colon V \to W$ and $S\colon W \to V$ be linear maps with $\dim V = \dim W < \infty$. Then

$$S \circ T = I_V \iff T \circ S = I_W.$$

Proof. ($\Rightarrow$) Assume $S \circ T = I_V$. Let $v \in \ker T$. Then

$$v = I_V(v) = (S \circ T)(v) = S(T(v)) = S(0_W) = 0_V.$$

Thus

$$\ker T = \{0_V\}$$

so T is injective.
Since $\dim V = \dim W < \infty$, T is invertible let $T^{-1}\colon W \to V$ be its inverse.
Then

$$\begin{aligned} S = S \circ I_W &= S \circ (T \circ T^{-1}) \\ &= (S \circ T) \circ T^{-1} \\ &= I_V \circ T^{-1} = T^{-1}. \end{aligned}$$

Hence

$$T \circ S = T \circ T^{-1} = I_W.$$

($\Leftarrow$) Repeat the proof with T and S swapped, and V and W swapped as well. □

8.3 Isomorphism

Definition 8.2

Let $T\colon V \to W$. T is called an *isomorphism* if T is invertible. In this case we write

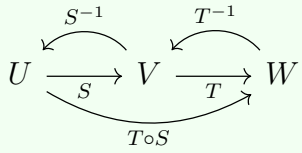
$$V \cong W,$$

and V and W are called *isomorphic*.

Theorem 8.5

Composition of isomorphisms is an isomorphism.
If $S\colon U \to V$ and $T\colon V \to W$ are isomorphisms, then $T \circ S\colon U \to W$ is an isomorphism.

S, T are isomorphisms.



$T \circ S\colon U \to W$ is an isomorphism.

Proof. Since S and T are isomorphisms, there exist

$$S^{-1}\colon V \to U, \qquad T^{-1}\colon W \to V.$$

Claim

$$S^{-1} \circ T^{-1} = (T \circ S)^{-1}.$$

Indeed,

$$\begin{aligned} T \circ S \circ (S^{-1} \circ T^{-1}) &= T \circ \underbrace{(S \circ S^{-1})}_{I_V} \circ T^{-1} \\ &= T \circ I_V \circ T^{-1} \\ &= T \circ T^{-1} \\ &= I_W \end{aligned}$$

and

$$\begin{aligned} (S^{-1} \circ T^{-1}) \circ (T \circ S) &= S^{-1} \circ \underbrace{(T^{-1} \circ T)}_{I_V} \circ S \\ &= S^{-1} \circ I_V \circ S \\ &= S^{-1} \circ S \\ &= I_U \end{aligned}$$

Thus $T \circ S$ is invertible with inverse $S^{-1} \circ T^{-1}$:

$$(T \circ S)^{-1} = S^{-1} \circ T^{-1}.$$

Therefore $T \circ S$ is an isomorphism. □

Lemma 8.6

Let V be an n-dimensional vector space over $\mathbb{F}$, and let $\{v_1, \ldots, v_n\}$ be a basis of V. Then there exists a unique isomorphism

$$\psi\colon \mathbb{F}^n \to V$$

such that

$$\psi(e_i) = v_i \qquad (i = 1, \ldots, n).$$

Proof. Define linear map

$$\psi(\alpha_1, \ldots, \alpha_n) = \alpha_1 v_1 + \cdots + \alpha_n v_n.$$

Since

$$span\{v_1, \ldots, v_n\} = V \implies \psi \text{ is surjective}$$

Since

$$\{v_1, \ldots, v_n\} \text{ is linearly independent} \implies \ker \psi = \{0\}$$

So ψ is injective. Hence ψ is an isomorphism. □

Theorem 8.7

Let V and W be finite-dimensional vector spaces over $\mathbb{F}$. Then

$$V \cong W \iff \dim V = \dim W.$$

Proof. ($\Rightarrow$) Suppose $V \cong W$, then there exists an invertible linear map $T\colon V \to W$, which is injective and surjective.

$$\begin{aligned} \dim V &= \dim \ker T + \dim \operatorname{Im} T \\ &= 0 + \dim W \\ &= \dim W. \end{aligned}$$

($\Leftarrow$) Suppose $\dim V = \dim W = n$. Choose basis for V and W

$$\begin{aligned} v_1, \ldots, v_n \\ w_1, \ldots, w_n \end{aligned}$$

By the lemma above, there are isomorphisms

$$\begin{aligned} \psi\colon \mathbb{F}^n &\to V \\ e_i &\mapsto v_i, \end{aligned}$$

and

$$\begin{aligned} \chi\colon \mathbb{F}^n &\to W \\ e_i &\mapsto w_i, \end{aligned}$$

for $i = 1, \ldots, n$. Then

$$\chi \circ \psi^{-1}\colon V \to W$$

is a composition of isomorphisms, hence an isomorphism.
Therefore

$$V \cong W$$

□

9 Lecture 9

From linear map to the matrices.

9.1 The space of matrices

Definition 9.1

Let V and W be finite-dimensional vector spaces over $\mathbb{F}$.

$$\mathbb{F}^{m,n} = \{m \times n \text{ matrices over } \mathbb{F}\}.$$

$(m \times n)$ matrices m #rows n #columns.

$\mathbb{F}^{m,n}$ is a vector space over $\mathbb{F}$.

$$[A_{ij}]_{\substack{i=1,\dots,m \\ j=1,\dots,n}} = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix}$$

$$[A_{ij}] + [B_{ij}] = [A_{ij} + B_{ij}]$$
$$\lambda[A_{ij}] = [\lambda A_{ij}]$$

Basis of $\mathbb{F}^{m,n}$ consists of matrices

$$E_{\substack{i=1,\dots,m \\ j=1,\dots,n}}$$

With a single entry 1 at the (i,j)-th position and the rest 0.

$$E_{ij} = \begin{bmatrix} 0 & \cdots & & \cdots & 0 \\ \vdots & & & & \vdots \\ & \cdots & 1 & \cdots & \\ \vdots & & & & \vdots \\ 0 & \cdots & & \cdots & 0 \end{bmatrix} \begin{matrix} \\ \\ i \\ \\ \\ \end{matrix}$$
$$\begin{matrix} & & j & & \end{matrix}$$

$$A = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} E_{ij}.$$

So the dimension of $\mathbb{F}^{m,n}$ is the number of different place in the matrix, which is mn.

$$\dim \mathbb{F}^{m,n} = mn$$

9.2 Matrix representation of linear maps

Definition 9.2

Matrix multiplication

$$\mathbb{F}^{m,n} \times \mathbb{F}^{n,p} \to \mathbb{F}^{m,p}$$
$$[A_{ij}] \times [B_{jk}] \mapsto [C_{ik}]$$
$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$
$$\begin{bmatrix} A_{11} & A_{12} & A_{13} & \cdots \\ \vdots & \ddots & \vdots & \ddots & \vdots \end{bmatrix} \times \begin{bmatrix} B_{11} & B_{12} & B_{13} & \cdots \\ \vdots & \ddots & \vdots & \ddots & \vdots \end{bmatrix}$$
$$C_{11} = A_{11}B_{11} + A_{12}B_{21} + A_{13}B_{31} + \cdots + A_{1n}B_{n1}$$

Theorem 9.1

$A \in \mathbb{F}^{m,n} \sim$ linear map $\mathbb{F}^n \to \mathbb{F}^m$

$$\mathbb{F}^n \to \mathbb{F}^m$$
$$x \mapsto Ax$$
$$\begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} A_{11}x_1 + \cdots + A_{1n}x_n \\ \vdots \\ A_{m1}x_1 + \cdots + A_{mn}x_n \end{bmatrix}$$

For vector spaces V and W

$$\dim V = n \quad \dim W = m$$

T is a linear map from V to W

$$V \xrightarrow{T} W$$

Choose basis

$$\{v_1, \ldots, v_n\} \quad \text{for } V$$
$$\{w_1, \ldots, w_m\} \quad \text{for } W$$
$$\mathbb{F}^n \xrightarrow[\psi]{\cong} V \qquad \mathbb{F}^m \xrightarrow[\psi]{\cong} W$$
$$e_i \mapsto v_i \qquad e_i \mapsto w_i.$$

The matrix representation of the linear map T with respect to the chosen bases is given by the following commutative diagram

$$\mathbb{F}^n \xrightarrow[\psi]{\cong} V \xrightarrow{T} W \xrightarrow[\psi^{-1}]{\cong} \mathbb{F}^m$$
$$\psi^{-1} \circ T \circ \psi$$

$$T(v_i) = \sum_{j=1}^{m} w_j A_{ji}$$

$$
\begin{aligned}
T\begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix} &= \begin{bmatrix} Tv_1 & \cdots & Tv_n \end{bmatrix} \\
&= \begin{bmatrix} w_1 & \cdots & w_m \end{bmatrix} \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
T\left(\sum_{i=1}^{n} x_i v_i\right) &= \sum_{i=1}^{n} x_i T v_i \\
&= \sum_{i=1}^{n} x_i \sum_{j=1}^{m} A_{ji} w_j \\
&= \sum_{j=1}^{m} w_j \left(\sum_{i=1}^{n} A_{ji} x_i\right)
\end{aligned}
$$

$\mathcal{M}(T)$ matrix which represents the linear map T with respect to the chosen bases.

Definition 9.3

The coordinate vector of a vector $w \in W$ with respect to the basis $\{w_1, \ldots, w_m\}$ is defined as follows:

$$
\begin{aligned}
\psi^{-1} &\colon W \to \mathbb{F}^n \\
w = \sum_{i=1}^{m} x_i w_i &\mapsto \begin{bmatrix} x_1 & \cdots & x_m \end{bmatrix}
\end{aligned}
$$

Coordinate vector of w with respect to the basis $\{w_1, \ldots, w_m\}$.

Example

$$\mathcal{D} \colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$$

$\{1, x, x^2, x^3\}$ basis of $\mathcal{P}_3(\mathbb{R})$ $\{1, x, x^2\}$ basis of $\mathcal{P}_2(\mathbb{R})$ Question: What is the matrix representation of the differentiation operator $\mathcal{D}$ with respect to the given bases?

$$
\mathcal{D}(1, x, x^2, x^3) = (0, 1, 2x, 3x^2) = (1, x, x^2) \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}
$$

$$
[\mathcal{M}(\mathcal{D})] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}
$$

9.3 The isomorphism $\mathcal{L}(V,W) \cong \mathbb{F}^{m\times n}$ and composition

Theorem 9.2

For vector spaces V and W, let $\dim V = n$, $\dim W = m$.

$$\begin{aligned} \mathcal{L}(V,W) &\to \mathbb{F}^{m\times n} \\ T &\mapsto \mathcal{M}(T) \end{aligned}$$

is a vector space isomorphism, and hence

$$\begin{aligned} \dim \mathcal{L}(V,W) &= mn \\ &= (\dim W)(\dim V) \end{aligned}$$

Proof. (From workshop $\mathcal{M}(T) = \psi^{-1} \circ T \circ \psi$)

$$\begin{aligned} \mathcal{M}(S+T) &= \psi^{-1} \circ (S+T) \circ \psi \\ &= \psi^{-1} \circ S \circ \psi + \psi^{-1} \circ T \circ \psi \\ &= \mathcal{M}(S) + \mathcal{M}(T) \end{aligned}$$

For $x \in \mathbb{F}^n$,

$$\begin{aligned} \psi^{-1} \underbrace{(S+T)(\psi(x))}_{S(\psi(x))+T(\psi(x))} &= \psi^{-1}(S(\psi(x)) + T(\psi(x))) \\ &= \psi^{-1}(S(\psi(x))) + \psi^{-1}(T(\psi(x))) \\ &= (\psi^{-1}S\psi + \psi^{-1}T\psi)(x) \end{aligned}$$

$$\begin{aligned} \mathcal{M}(\lambda T) &= \psi^{-1} \circ (\lambda T) \circ \psi \\ &= \lambda(\psi^{-1} \circ T \circ \psi) \\ &= \lambda \mathcal{M}(T) \end{aligned}$$

$\mathcal{M}$ is a linear map.

$$\begin{aligned} \mathcal{M}(T) &= \psi^{-1} \circ T \circ \psi \\ \psi \circ \mathcal{M}(T) \circ \psi^{-1} &= \underline{\psi \circ \psi^{-1}} \circ T \circ \underline{\psi \circ \psi^{-1}} \\ &= T \end{aligned}$$

□

Theorem 9.3

Composition of linear maps is represented by matrix multiplication with respect to the chosen bases.

Proof. U, V, W are vector spaces over the field $\mathbb{F}$.

$$\dim U = p, \quad \dim V = n, \quad \dim W = m.$$

Let $S\colon U \to V$ and $T\colon W \to U$ be linear maps with chosen bases for V, W, U. Then the composition $S \circ T\colon W \to V$ is represented as follows

$$W \xrightarrow{T} U \xrightarrow{S} V, \qquad W \xrightarrow{S\circ T} V$$

Chosen basis

$$U\colon u_1, \ldots, u_p$$
$$W\colon w_1, \ldots, w_m$$
$$V\colon v_1, \ldots, v_n$$

$$Su_i = \sum_{j=1}^{n} v_j S_{ji} \qquad Tw_k = \sum_{i=1}^{p} u_i T_{ik}$$

$$(S \circ T)w_k = \sum_{j=1}^{n} v_j C_{jk}$$

where

$$C_{jk} = \sum_{i=1}^{p} S_{ji} T_{ik} \Leftrightarrow \mathcal{M}(S \circ T) = \mathcal{M}(S)\mathcal{M}(T)$$

Then

$$\begin{aligned} S(Tw_k) &= S\left(\sum_{i=1}^{p} u_i T_{ik}\right) \\ &= \sum_{i=1}^{p} T_{ik} S(u_i) \\ &= \sum_{i=1}^{p} T_{ik} \left(\sum_{j=1}^{n} v_j S_{ji}\right) \\ &= \sum_{j=1}^{n} v_j \left(\sum_{i=1}^{p} S_{ji} T_{ik}\right) \\ &= \sum_{j=1}^{n} v_j C_{jk} \end{aligned}$$

□

9.4 Normal Form for Matrices

Normal form for matrices representing linear maps (cf. proof of the Rank theorem)
For linear map T

$$T\colon V \to W$$

Choose basis for $\ker T$

$$u_1, \ldots, u_k$$

Complete it to a basis for the whole space V

$$v_1, \ldots, v_r, u_1, \ldots, u_n$$

So basis for the $\operatorname{im} T \subseteq W$ is

$$Tv_1, \ldots, Tv_r$$

Complete it to a basis for the space W

$$Tv_1, \ldots, Tv_r, w_1, \ldots, w_p$$

apply the linear map T to the basis vectors of V

$$\begin{aligned} T(v_1,\dots,v_r,u_1,\dots,u_k) &= (Tv_1,\dots,Tv_r,Tu_1,\dots,Tu_k) \\ &= (Tv_1,\dots,Tv_r,0,\dots,0) \\ &= (Tv_1,\dots,Tv_r,w_1,\dots,w_p) \end{aligned}$$

Since

$$\forall i = 1,\dots,k \quad Tu_i = 0$$

So if we choose the bases for V and W as described above, the matrix representation of T with respect to these bases is, which called the *normal form* of the matrix representation of T.

$$\mathcal{M}(T) = \begin{array}{c} \begin{matrix} \overset{r}{} & & \overset{n}{} \end{matrix} \\ \left[\begin{array}{ccc|c} 1 & \cdots & 0 & \\ \vdots & \ddots & \vdots & 0 \\ 0 & \cdots & 1 & \\ \hline & 0 & & 0 \end{array}\right] \begin{matrix} r \\ \\ \\ p \end{matrix} \end{array}$$

$r = \operatorname{rank} T = \dim \operatorname{im} T$

$$\mathcal{M}(T) = \left(\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array}\right)$$

Remark

Careat: For a linear map

$$T\colon V \to W$$

The poblem will be much harder.

---

cf. Jordan Normal Form

10 Lecture 10

10.1 Change of Basis

$T : V \to W$
Let V and W be vector spaces over $\mathbb{F}$ and $\dim V = n$, $\dim W = m$.

$$\begin{cases} v_1,\dots,v_n \\ w_1,\dots,w_m \end{cases} \quad \text{bases for } V, W$$

Define the isomorphisms

$$\begin{array}{cc} \mathbb{F}^n \underset{\varphi}{\overset{\cong}{\to}} V & \mathbb{F}^m \underset{\psi}{\overset{\cong}{\to}} W \\ e_i \mapsto v_i & e_i \mapsto w_i \end{array}$$

We have the following commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ {\scriptstyle\varphi}\uparrow & & \downarrow{\scriptstyle\psi^{-1}} \\ \mathbb{F}^n & \xrightarrow[m(T)]{} & \mathbb{F}^m \end{array}$$

$$m(T) = \psi^{-1} T \varphi$$

$$Tv_i = \sum_{j=1}^{m} w_j T_{ji}$$

10.2 Change of Basis for Linear Operators

10.2.1 Change of Basis Matrices

For a finite dimensional vector space V over $\mathbb{F}$ ($\dim V = n$)

$$\begin{cases} v_1, \ldots, v_n \\ v_1', \ldots, v_n' \end{cases} \quad \text{bases for } V$$

Then for $v \in V$

$$v = \sum_{i=1}^{n} a_i v_i = \sum_{i=1}^{n} a_i' v_i'$$

$$v_i = \sum_{j=1}^{n} v_j' T_{ji}$$

$$v_i' = \sum_{j=1}^{n} v_j S_{ji}$$

T_{ji}, S_{ji} are the entries of $n \times n$ matrices T & S

$$T = \begin{pmatrix} T_{11} & \cdots & T_{1n} \\ \vdots & \ddots & \vdots \\ T_{n1} & \cdots & T_{nn} \end{pmatrix} \quad S = \begin{pmatrix} S_{11} & \cdots & S_{1n} \\ \vdots & \ddots & \vdots \\ S_{n1} & \cdots & S_{nn} \end{pmatrix}$$

$$\begin{aligned} v_i &= \sum_{j=1}^{n} v_j' T_{ji} \\ &= \sum_{j=1}^{n} \left(\sum_{k=1}^{n} v_k S_{kj} \right) T_{ji} \\ &= \sum_{k=1}^{n} v_k \left(\sum_{j=1}^{n} S_{kj} T_{ji} \right) \end{aligned}$$

Linear independence of $\{v_1, \ldots, v_n\}$ implies that

$$\underbrace{\sum_{j=1}^{n} S_{kj} T_{ji}}_{\text{(ki) entry of } ST} = \begin{cases} 1 & k = i \\ 0 & k \neq i \end{cases}$$

For $n \times n$ matrices over $\mathbb{F}$, the above shows that S is the inverse of T.

$$T, S \colon \mathbb{F}^n \to \mathbb{F}^n$$

$$TS = I \Leftrightarrow ST = I$$

And this shows that T is invertible with inverse S.

$$I = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{pmatrix}$$

Remark

Upshot

$$(v_1', \ldots, v_n') = (v_1, \ldots, v_n) \begin{pmatrix} S_{11} & \cdots & S_{1n} \\ \vdots & \ddots & \vdots \\ S_{n1} & \cdots & S_{nn} \end{pmatrix} \quad \& \quad T = S^{-1}$$

$$(v_1, \ldots, v_n) = (v_1', \ldots, v_n') \begin{pmatrix} T_{11} & \cdots & T_{1n} \\ \vdots & \ddots & \vdots \\ T_{n1} & \cdots & T_{nn} \end{pmatrix}$$

S is the change of basis matrix from

$$v_1, \ldots, v_n \text{ to } v_1', \ldots, v_n'$$

Definition 10.1

Group (for linear algebra)

1. A set with a multiplication (associative)
2. $\exists$ a multiplicative identity

Remark

Invertible $n \times n$ matrices form a group

Here the multiplication is the product of matrices & the identity is I.
It is called the General linear group $GL(n, \mathbb{F})$, which is the set of all invertible $n \times n$ matrices over $\mathbb{F}$.

10.2.2 Transformation of Matrix Representation

Consider $T : V \to V$

$$\begin{cases} v_1, \ldots, v_n \\ v_1', \ldots, v_n' \end{cases}$$

Are related by a matrix $S \in GL(n, \mathbb{F})$

$$(v_1', \ldots, v_n') = (v_1, \ldots, v_n)S$$

We have the relationship between the coordinates of v_i and v_i'.

$$Tv_i = \sum_{j=1}^{n} v_j T_{ji} \qquad [T_{ij}] \in \mathbb{F}^{n \times n}$$

$$Tv'_i = \sum_{j=1}^{n} v'_j T'_{ji} \qquad [T'_{ij}] \in \mathbb{F}^{n\times n}$$

We can express the action of T on the basis vectors in terms of matrices

$$\begin{aligned} T(v_1,\ldots,v_n) &= (Tv_1,\ldots,Tv_n) = (v_1,\ldots,v_n)\mathcal{M}(T) \\ T(v'_1,\ldots,v'_n) &= (Tv'_1,\ldots,Tv'_n) = (v'_1,\ldots,v'_n)\mathcal{M}'(T) \\ &= (v_1,\ldots,v_n)S\mathcal{M}'(T) \end{aligned}$$

$$\begin{aligned} T(v'_1,\ldots,v'_n) &= T(v_1,\ldots,v_n)S \\ &= (v_1,\ldots,v_n)\mathcal{M}(T)S \end{aligned}$$

So we have

$$(v_1,\ldots,v_n)\mathcal{M}(T)S = (v_1,\ldots,v_n)S\mathcal{M}'(T)$$

So that

$$\mathcal{M}(T)S = S\mathcal{M}'(T)$$

Right multiplication by S^{-1}

$$\mathcal{M}'(T) = S^{-1}\mathcal{M}(T)S$$

$$Tv_i = \sum_{j=1}^{n} v_j T_{ji}$$

For the change of basis $v'_i = \sum_{j=1}^{n} v_j S_{ji}$ we have

$$\begin{aligned} Tv'_i &= \sum_{j=1}^{n} v'_j T'_{ji} \\ &= \sum_{j=1}^{n} \left(\sum_{k=1}^{n} v_k S_{kj} \right) T'_{ji} \\ &= \sum_{k=1}^{n} v_k \left(\sum_{j=1}^{n} S_{kj} T'_{ji} \right) \end{aligned}$$

Then we can also express Tv'_i in terms of Tv_j and the change of basis matrix S.

$$\begin{aligned} Tv'_i &= T\left(\sum_{j=1}^{n} v_j S_{ji} \right) \\ &= \sum_{j=1}^{n} (Tv_j) S_{ji} \\ &= \sum_{j=1}^{n} \left(\sum_{k=1}^{n} v_k \mathcal{M}(T)_{kj} \right) S_{ji} \\ &= \sum_{k=1}^{n} v_k \left(\sum_{j=1}^{n} \mathcal{M}(T)_{kj} S_{ji} \right) \end{aligned}$$

So we have

$$\sum_{j=1}^{n} S_{kj}\mathcal{M}'(T)_{ji} = \sum_{j=1}^{n} \mathcal{M}(T)_{kj} S_{ji}$$

$$(S\mathcal{M}'(T))_{ki} = (\mathcal{M}(T)S)_{ki}$$
$$S\mathcal{M}'(T) = \mathcal{M}(T)S$$

Right multiplication by S^{-1} gives

$$\mathcal{M}'(T) = S^{-1}\mathcal{M}(T)S$$

10.2.3 Summary

For a linear map T

$$T\colon V \to V \quad (\dim V = n)$$

And two different bases for V

$$\begin{cases} v_1, \dots, v_n \\ v_1', \dots, v_n' \end{cases}$$

We have two different matrix representations of T corresponding to the two bases.

$$\begin{aligned} &\mathcal{M}(T) && \text{rel. to } v_1, \dots, v_n \\ &\mathcal{M}'(T) && \text{rel. to } v_1', \dots, v_n' \end{aligned}$$

Then

$$\mathcal{M}'(T) = S^{-1}\mathcal{M}(T)S$$

Where

1. $S \in GL(n, \mathbb{F})$
2. $(v_1', \dots, v_n') = (v_1, \dots, v_n)S$

> **Remark**
>
> Any intrinsic properties of $T\colon V \to V$ must be invariant under change of basis.
>
> - trace of T
> - determinant of T
> - characteristic poly of T
> - $\dots$

10.3 Change of Basis for $T\colon V \to W$

For a linear map

$$T\colon V \to W$$
$$\dim V = n \qquad \dim W = m$$

Choose bases of V and W

$$\left.\begin{matrix} v_1, \dots, v_n \text{ for } V \\ w_1, \dots, w_m \text{ for } W \end{matrix}\right\} \implies \mathcal{M}(T) \in \mathbb{F}^{m,n}$$

$$Tv_i = \sum_{j=1}^{m} w_j T_{ji} \leftarrow \text{ entries of } \mathcal{M}(T)$$

Choose other bases of V and W

$$\left.\begin{array}{ll} v'_1, \ldots, v'_n & \text{for } V \\ w'_1, \ldots, w'_m & \text{for } W \end{array}\right\} \implies \mathcal{M}'(T) \in \mathbb{F}^{m,n}$$

$$Tv'_i = \sum_{j=1}^{m} w'_j \mathcal{M}'(T)_{ji} \leftarrow \text{ entries of } \mathcal{M}'(T)$$

Two change-of-basis matrices

$$\begin{array}{ll} S \in GL(n, \mathbb{F}) & (v'_1, \ldots, v'_n) = (v_1, \ldots, v_n)S \\ R \in GL(m, \mathbb{F}) & (w'_1, \ldots, w'_m) = (w_1, \ldots, w_m)R \end{array}$$

Guess:

$$\underset{m'(T)}{\left(m \times n\right)} = \underset{R^{-1}}{\left(m \times m\right)} \underset{m(T)}{\left(\begin{array}{c}\text{rows} \times \text{columns} \\ m \quad\quad n\end{array}\right)} \underset{S}{\left(n \times n\right)}$$

$$\mathcal{M}'(T) = R^{-1}\mathcal{M}(T)S$$

Then we can verify this by direct computation

$$\begin{aligned} T(v'_1, \ldots, v'_n) = (Tv'_1, \ldots, Tv'_n) &= (w'_1, \ldots, w'_m)\mathcal{M}'(T) \\ &= (w_1, \ldots, w_m)R\mathcal{M}'(T) \end{aligned}$$

$$\begin{aligned} T(v'_1, \ldots, v'_n) &= T(v_1, \ldots, v_n)S \\ &= (w_1, \ldots, w_m)\mathcal{M}(T)S \end{aligned}$$

So we have

$$(w_1, \ldots, w_m)\mathcal{M}(T)S = (w_1, \ldots, w_m)R\mathcal{M}'(T)$$
$$\mathcal{M}(T)S = R\mathcal{M}'(T)$$

Right multiplication by R^{-1} gives

$$\mathcal{M}'(T) = R^{-1}\mathcal{M}(T)S$$

Remark

Last lecture:

$$\exists R \in GL(m, \mathbb{F})$$
$$\exists S \in GL(n, \mathbb{F})$$

$$\mathcal{M}'(T) = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} \quad r = \operatorname{rank} T = \dim \operatorname{im} T$$

11 Lecture 11

- Product
- Quotient

Vector spaces are sets of structures

11.1 Product

11.1.1 Definition and Basic Properties

Definition 11.1

V_1 and V_2 are vector spaces over $\mathbb{F}$. The *product* of V_1 and V_2 is the vector space

$$V_1 \times V_2 = \{(v_1, v_2) \mid v_1 \in V_1, v_2 \in V_2\}$$

with addition and scalar multiplication
for $v_1, v_2 \in V_1 \times V_2$ and $\alpha \in \mathbb{F}$

$$(v_1, v_2) + (v_1', v_2') = (v_1 + v_1', v_2 + v_2')$$
$$\alpha(v_1, v_2) = (\alpha v_1, \alpha v_2)$$

Remark

The dimension of the product of two vector spaces is the sum of their dimensions.

$$\dim V_1 = n_1$$
$$\dim V_2 = n_2$$

Claim:

$$\dim(V_1 \times V_2) = n_1 + n_2$$

Proof. Choose

$$\{e_1, \ldots, e_{n_1}\} \text{ is a basis of } V_1$$
$$\{f_1, \ldots, f_{n_2}\} \text{ is a basis of } V_2$$

Then

$$(e_1, 0), \ldots, (e_{n_1}, 0), (0, f_1), \ldots, (0, f_{n_2})$$

is a basis of $V_1 \times V_2$. For $(v_1, v_2) \in V_1 \times V_2$, we can write

$$v_1 = \sum_{i=1}^{n_1} \alpha_i e_i, \quad v_2 = \sum_{j=1}^{n_2} \beta_j f_j, \quad \alpha_i, \beta_j \in \mathbb{F}$$

$$(v_1, v_2) = \sum_{i=1}^{n_1} \alpha_i (e_i, 0) + \sum_{j=1}^{n_2} \beta_j (0, f_j)$$

for some $\alpha_i, \beta_j \in \mathbb{F}$, showing that the set spans $V_1 \times V_2$. Linear independence is straightforward to verify.

$$\text{If } \sum_{i=1}^{n_1} \alpha_i (e_i, 0) + \sum_{j=1}^{n_2} \beta_j (0, f_j) = (0, 0)$$

Then

$$\sum_{i=1}^{n_1} \alpha_i e_i = 0 \text{ and } \sum_{j=1}^{n_2} \beta_j f_j = 0$$

Then all the coefficients α_i and β_j must be zero, proving linear independence. □

Definition 11.2

$V_1, \ldots, V_m$ are vector spaces over $\mathbb{F}$. The *product* of $V_1, \ldots, V_m$ is the vector space over $\mathbb{F}$ defined by

$$V_1 \times \cdots \times V_m = \{(v_1, \ldots, v_m) \mid v_i \in V_i \text{ for } i = 1, \ldots, m\}$$

The operations on $V_1 \times \cdots \times V_m$ are defined componentwise

$$(v_1, \ldots, v_m) + (v_1', \ldots, v_m') = (v_1 + v_1', \ldots, v_m + v_m')$$

$$\alpha(v_1, \ldots, v_m) = (\alpha v_1, \ldots, \alpha v_m)$$

Theorem 11.1

$$\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m$$

Proof.

$$\begin{aligned} &e_1^{(1)}, \ldots, e_{n_1}^{(1)} && \text{is a basis of } V_1 \\ &e_1^{(2)}, \ldots, e_{n_2}^{(2)} && \text{is a basis of } V_2 \\ &\vdots \\ &e_1^{(m)}, \ldots, e_{n_m}^{(m)} && \text{is a basis of } V_m \end{aligned}$$

Then

$$\begin{aligned} &(e_j^{(1)}, 0, \ldots, 0) && j = 1, \ldots, n_1 \\ &(0, e_j^{(2)}, \ldots, 0) && j = 1, \ldots, n_2 \\ &\vdots \\ &(0, \ldots, 0, e_j^{(m)}) && j = 1, \ldots, n_m \end{aligned}$$

is a basis of $V_1 \times \cdots \times V_m$, proving that

$$\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m$$

□

11.1.2 Relation to Direct Sum

Relate the product to the direct sum

Theorem 11.2

Let $V_1, \ldots, V_m$ are subspaces of a vector space V. We can define the linear map

$$\begin{aligned} \phi : V_1 \times \cdots \times V_m &\to V \\ (v_1, \ldots, v_m) &\mapsto v_1 + \cdots + v_m \end{aligned}$$

The sum $V_1 + \cdots + V_m$ is direct $\iff$ ϕ is an isomorphism.

Remark

Direct

$$v_1 + \cdots + v_m = 0 \iff v_1 = \cdots = v_m = 0$$

for all $v_i \in V_i$.

Proof. Proof needed for both directions of the equivalence.

($\Longrightarrow$) Suppose $\phi(v_1, \ldots, v_m) = 0$

1. ϕ is injective
   (by definition of direct sum)
2. ϕ is surjective
   (by construction)

$\Longrightarrow$ ϕ is an isomorphism.

($\Longleftarrow$) Suppose ϕ is an isomorphism. Then ϕ is injective

$$\phi(v_1, \ldots, v_m) = 0 \Longrightarrow (v_1, \ldots, v_m) = 0$$

Hence

$$v_1 + \cdots + v_m = 0 \Longrightarrow v_1 = \cdots = v_m = 0$$

for all $v_i \in V_i$, proving that the sum is direct. □

Theorem 11.3

Corollary

$$\dim(V_1 \oplus \cdots \oplus V_m) = \sum_{i=1}^{m} \dim V_i$$

11.2 Quotient

11.2.1 Equivalence Relation

Definition 11.3

Define an equivalence relation on V:

$$v_1 \sim v_2 \iff v_1 - v_2 \in U.$$

an equivalence relation satisfying reflexivity, symmetry, and transitivity.

1. $v \sim v$
$$v - v = 0 \in U$$
2. If $v_1 \sim v_2$, then $v_2 \sim v_1$
$$v_2 - v_1 = -(v_1 - v_2) \in U v_1 - v_2 \in U$$
3. If $v_1 \sim v_2$ and $v_2 \sim v_3$, then $v_1 \sim v_3$
$$v_1 - v_2 \in U v_2 - v_3 \in U$$
So
$$v_1 - v_3 = (v_1 - v_2) + (v_2 - v_3) \in U$$
$$v_1 - v_3 \in U$$

Definition 11.4

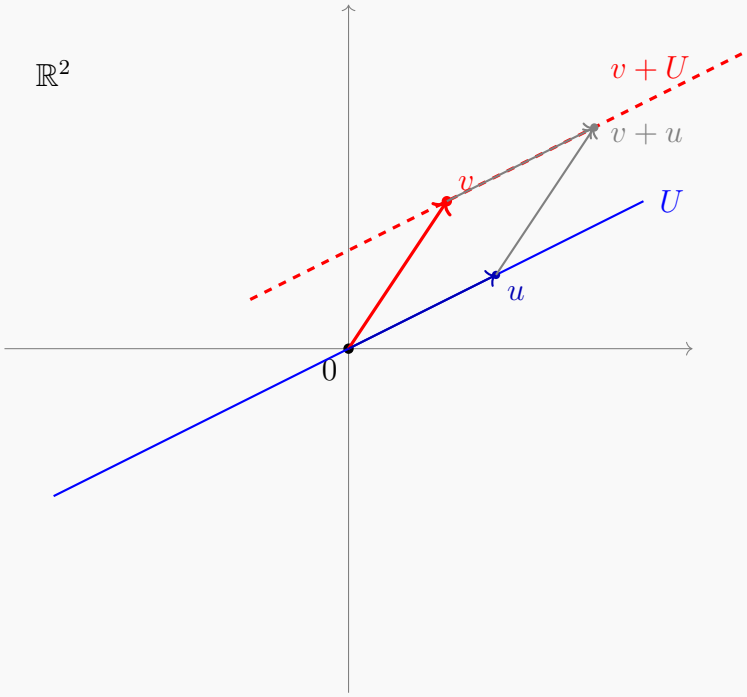
The equivalence class of a vector $v \in V$ under the relation $\sim$ is the set

$$[v] = \{w \in V \mid w \sim v\} = v + U.$$

is the translate of the subspace U by the vector v.

Example

Let $V = \mathbb{R}^2$ and U be a subspace.



11.2.2 The quotient

Definition 11.5

The quotient space V/U is the set of equivalence classes.

Let V be a vector space and U a subspace of V. The quotient space V/U is the set of cosets of U in V:

$$V/U = \{v + U \mid v \in V\}.$$

Definition 11.6

The quotient map

The quotient map is the map

$$\begin{aligned} \pi \colon V &\to V/U \\ v &\mapsto [v] = v + U \end{aligned}$$

Theorem 11.4

Claim V/U can be made into a vector space such that π is linear

$$[v] = v + U$$
$$[w] = w + U$$

For $\lambda \in \mathbb{F}$

$$[v] + [w] = [v + w]$$
$$\lambda[v] = [\lambda v]$$

Proof. We need to show two things to verify that this definition of addition and scalar multiplication on V/U are proper.

1. *Operation well defined*

   First, we need to check that the operations are well defined

   $$\begin{aligned} [v] = [v'] &\implies v - v' \in U \\ &\implies \exists u \in U \text{ s.t. } v - v' = u \\ &\implies v = v' + u \end{aligned}$$

   So

   $$[\lambda v] = [\lambda(v' + u)] = [\lambda v' + \lambda u] = [\lambda v']$$

   We have

   $$\lambda v \sim \lambda v' \implies [\lambda v] = [\lambda v']$$

   Similarly, addition is well defined.

   $$\begin{aligned} [v] = [v'] &\implies v - v' \in U \\ [w] = [w'] &\implies w - w' \in U \\ &\implies (v + w) - (v' + w') = (v - v') + (w - w') \in U \\ &\implies [v + w] = [v' + w'] \end{aligned}$$

2. *π is linear*

   Then we check

   $$\pi \colon V \to V/U$$
   $$v \mapsto [v]$$

   is linear Let $v, w \in V$ and $\lambda \in \mathbb{F}$. Then

   $$\pi(v + w) = [v + w] = [v] + [w] = \pi(v) + \pi(w),$$

   and

   $$\pi(\lambda v) = [\lambda v] = \lambda[v] = \lambda\pi(v).$$

   Hence, π is linear.

□

Remark

well-defined is the operation being independent of the representative chosen.

Lemma 11.5

The canonical projection $\pi\colon V \to V/U$ is surjective.

Proof. Let

$$[v] \in V/U$$

By definition of the quotient space

$$[v] = v + U$$

for some $v \in V$. Then

$$\pi(v) = [v]$$

showing that every element of V/U is in the image of π.

$$\operatorname{im}(\pi) = V/U$$

□

Lemma 11.6

The kernel of the canonical projection $\pi\colon V \to V/U$ is U.

Proof. Let $v \in V$. Then

$$\begin{aligned} v \in \ker(\pi) &= \{\ v \in V \mid \pi(v) = [0]\ \} \\ &= \{\ v \in V \mid [v] = [0]\ \} \\ &= U \end{aligned}$$

So that we have

$$v - 0 \in U \iff v \in U$$

Hence

$$\ker(\pi) = U$$

□

Theorem 11.7

V is finite dimensional

$$\dim V/U = \dim V - \dim U$$

Proof. Use rank theorem since π is linear, we have

$$\dim V = \dim(\ker(\pi)) + \dim(\operatorname{im}(\pi))$$

And

$$\begin{aligned} \ker(\pi) &= U \\ \operatorname{im}(\pi) &= V/U \end{aligned}$$

So we have

$$\dim V = \dim U + \dim V/U \implies \dim V/U = \dim V - \dim U$$

□

12 Lecture 12

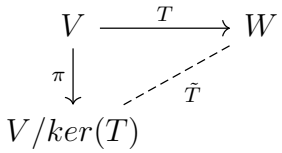
Dual spaces and annihilators

12.1 Last Lecture

For $U \subseteq V$

$$\begin{aligned} V &\xrightarrow{\pi} V/U \\ v &\mapsto [v] = v + U \end{aligned}$$

We have the following commutative diagram



$U = \ker(T) \subseteq V$ The next we have the definition of the induced map

Definition 12.1

$$\tilde{T} \colon V/\ker(T) \to W$$

defined by

$$\tilde{T}([v]) = T(v)$$

for all $v \in V$, is called the *induced map* or *quotient map* of T by $\ker(T)$.

Lemma 12.1

The kernel and image of the induced map $\tilde{T}$ are given by

$$\begin{cases} \ker(\tilde{T}) &= \{[0]\} \\ \operatorname{im}(\tilde{T}) &= \operatorname{im}(T) \end{cases}$$

Proof. For the kernel, we have

$$\begin{aligned} \ker(\tilde{T}) &= \{[v] \in V/\ker(T) \mid \tilde{T}([v]) = 0\} \\ &= \{[v] \in V/\ker(T) \mid T(v) = 0\} \\ &= \ker(T)/\ker(T) \\ &= \{[0]\} \end{aligned}$$

For the image, we have

$$\begin{aligned} \operatorname{im}(\tilde{T}) &= \{\tilde{T}([v]) \mid [v] \in V/\ker(T)\} \\ &= \{T(v) \mid v \in V\} \\ &= \operatorname{im}(T) \end{aligned}$$

□

12.2 Dual Spaces

12.2.1 Dual Space and Dual Basis

Let V and W be finite dimensional vector spaces over a field $\mathbb{F}$.

V, W	$\mathbb{F}$	finite dimensional
$\mathcal{L}(V, W)$	$\mathbb{F}$	$\dim \mathcal{L}(V, W) = \dim V \dim W$
V'	$\mathbb{F}$	$\dim V' = \dim V$

For the spaces above, when $W = \mathbb{F}$, we write $V' = \mathcal{L}(V, \mathbb{F})$ for the dual space of V.

Definition 12.2

The dual space of a vector space V over a field $\mathbb{F}$ is the vector space $V' = \mathcal{L}(V, \mathbb{F})$ consisting of all linear maps from V to $\mathbb{F}$.

Definition 12.3

Let $v_1, \ldots, v_n$ be a basis for V. The *(Canonical) dual basis* of V' corresponding to the basis $v_1, \ldots, v_n$ is $\varphi_1, \ldots, \varphi_n \in V'$ defined by

$$\varphi_i(v_j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

So, for any vector $v \in V$, we can express it in terms of the basis $v_1, \ldots, v_n$ as

$$v = a_1 v_1 + \cdots + a_n v_n, \quad a_i \in \mathbb{F}$$

$$\varphi_i(v) = a_1 \varphi_i(v_1) + \cdots + a_n \varphi_i(v_n) = a_i$$

$$v = \varphi_1(v) v_1 + \cdots + \varphi_n(v) v_n$$

Theorem 12.2

The canonical dual basis

$$\varphi_1, \ldots, \varphi_n$$

is a basis for V'.

Proof.

$$\dim V' = \dim V = n \quad \& \quad \varphi_1, \ldots, \varphi_n$$

This follows from linear independence.

$$a_1 \varphi_1 + \cdots + a_n \varphi_n = 0$$

$$\iff \forall v \in V \ (a_1 \varphi_1 + \cdots + a_n \varphi_n)(v) = 0$$

Then $\forall i$

$$0 = (a_1 \varphi_1 + \cdots + a_n \varphi_n)(v_i) = a_i$$

□

12.2.2 Dual map

For a linear map $T : V \to W$ and a linear functional $\varphi \in W'$, the composition $\varphi \circ T$ is a linear functional in V'. We have the following commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ & \searrow^{\varphi\circ T} & \downarrow{\scriptstyle \varphi} \\ & & \mathbb{F} \end{array}$$

So that we can define the dual map from W' to V'

Definition 12.4

For a linear map $T : V \to W$, the dual map $T' \colon W' \to V'$ is defined by

$$\begin{aligned} T' \colon W' &\to V' \\ \varphi &\mapsto T'(\varphi) = \varphi \circ T \end{aligned}$$

Which is

$$W' \xrightarrow{T'} V'$$

Lemma 12.3

The dual map T' is linear.

Proof. $\forall \varphi, \psi \in W', \forall \lambda \in \mathbb{F}$

$$\begin{aligned} T'(\varphi + \lambda\psi)(v) &= (\varphi + \lambda\psi)(Tv) \\ &= \varphi(Tv) + \lambda\psi(Tv) \\ &= (T'\varphi)(v) + \lambda(T'\psi)(v) \\ &= (T'\varphi + \lambda T'\psi)(v). \end{aligned}$$

Hence

$$T'(\varphi + \lambda\psi) = T'\varphi + \lambda T'\psi$$

linearity □

Example

Consider the differentiation operator and evaluation at a point $x_0 \in \mathbb{R}$ which is a linear functional.

$$\begin{aligned} \mathcal{D} \colon \mathcal{P}(\mathbb{R}) &\to \mathcal{P}(\mathbb{R}) \\ p(x) &\mapsto p'(x) \end{aligned}$$

$$\begin{aligned} \varphi \colon \mathcal{P}(\mathbb{R}) &\to \mathbb{R} \\ p(x) &\mapsto p(x_0) \end{aligned}$$

We have

$$(\mathcal{D}\varphi)(p) = \varphi(\mathcal{D}p) = \varphi(p') = p'(x_0)$$

Define $\varphi(p) = \int_0^1 p(x)dx$

$$(\mathcal{D}\varphi)(p) = \varphi(p') = \int_0^1 p'(x)dx = p(1) - p(0)$$

12.2.3 Linearity of Dual Operation

For linear maps T

$$V \xrightarrow{T} W \qquad W' \xrightarrow{T'} V'$$

Their action on dual maps is given by

$$\begin{aligned} \mathcal{L}(V,W) &\to \mathcal{L}(W',V') \\ T &\mapsto T' \end{aligned}$$

Which is a linear map from $\mathcal{L}(V,W)$ to $\mathcal{L}(W',V')$.

Definition 12.5

The dual operation $T \mapsto T'$ is defined for linear maps $T \in \mathcal{L}(V,W)$ as the map $T' \in \mathcal{L}(W',V')$ given by

$$(T'\varphi)(v) = \varphi(Tv)$$

Lemma 12.4

The dual operation

$$T \mapsto T'$$

is linear.

Proof. For $S, T \in \mathcal{L}(V,W)$ and $\lambda \in \mathbb{F}$, Check

$$(S+\lambda T)' = S' + \lambda T'$$

For any $v \in V$ and $\varphi \in W'$,

$$\begin{aligned} ((S+\lambda T)'\varphi)(v) &= \varphi((S+\lambda T)(v)) \\ &= \varphi(Sv + \lambda Tv) \\ &= \varphi(Sv) + \lambda\varphi(Tv) \\ &= (S'\varphi + \lambda T'\varphi)(v) \end{aligned}$$

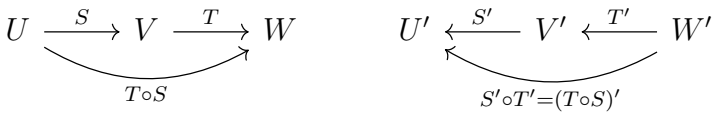
So that

$$(S+\lambda T)' = S' + \lambda T'$$

□

12.2.4 Composition of Dual Maps

For linear maps $S\colon U \to V$ and $T\colon V \to W$, we have following diagrams



Theorem 12.5

For the composition of dual maps, we have

$$(T \circ S)' = S' \circ T'$$

Proof. For any $\varphi \in W'$

$$\begin{aligned}(S' \circ T')(\varphi) = S'(T'\varphi) &= (T'\varphi) \circ S \\ &= (\varphi \circ T) \circ S \\ &= \varphi \circ (T \circ S) \\ &= (T \circ S)'(\varphi)\end{aligned}$$

□

Notation

The composition of dual maps reverses the order of composition

$$(T \circ S)' = S' \circ T'$$

Has such similarity to the transposition of matrices

$$(AB)^T = B^T A^T$$

And in fact, through choosing appropriate bases, the dual map corresponds to the transpose of the matrix representing the original linear map.

$$T \colon V \to W \qquad \begin{cases} v_1, \dots, v_n \\ w_1, \dots, w_m \end{cases} \qquad \mathcal{M}(T)$$

$$T' \colon W' \to V' \qquad \begin{cases} \psi_1, \dots, \psi_m \\ \varphi_1, \dots, \varphi_n \end{cases} \qquad \mathcal{M}(T') = \mathcal{M}(T)^t$$

12.3 Annihilators

Definition 12.6

Let V be a vector space over a field $\mathbb{F}$ and

$$U \subseteq V$$

The *annihilator* of U in V' is defined as

$$U^0 = \{\varphi \in V' \mid \varphi(u) = 0 \ \forall u \in U\}$$

Example

$V = \mathbb{R}^3$ and let

$$U = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\}$$

Then $U^0 = ?$

Let $\varphi_1, \varphi_2, \varphi_3$ and e_1, e_2, e_3 be the canonical dual basis of $(\mathbb{R}^3)'$ and basis of $\mathbb{R}^3$.

$$(x_1, x_2, x_3) = x_1 e_1 + x_2 e_2 + x_3 e_3$$

Let $\varphi \in (\mathbb{R}^3)'$

$$\varphi = a_1\varphi_1 + a_2\varphi_2 + a_3\varphi_3$$

Then

$$\varphi \in U^0 \Leftrightarrow \varphi(x_1, x_2, -x_1 - x_2) = 0$$

$$a_1 x_1 + a_2 x_2 + a_3(-x_1 - x_2) = 0$$

$$(a_1 - a_3)x_1 + (a_2 - a_3)x_2 = 0$$

So we must have

$$a_1 - a_3 = 0$$

$$a_2 - a_3 = 0$$

$$a_1 = a_2 = a_3$$

$$\varphi = a_1(\varphi_1 + \varphi_2 + \varphi_3)$$

Therefore

$$U^0 = \{a(\varphi_1 + \varphi_2 + \varphi_3) \mid a \in \mathbb{R}\}$$

Theorem 12.6

The annihilator which is

$$U^0 \subseteq V'$$

is a subspace of V'

Proof. Let $\varphi, \psi \in U^0$ and $\lambda \in \mathbb{F}$

$$\varphi + \lambda\psi \overset{?}{\in} U^0$$

Let $u \in U$

$$(\varphi + \lambda\psi)(u) = \overset{=0}{\varphi(u)} + \lambda\overset{=0}{\psi(u)} = 0$$

So indeed

$$\varphi + \lambda\psi \in U^0$$

□

Theorem 12.7

For subspace $U \subseteq V$ of a finite-dimensional vector space V $(\dim V < \infty)$

$$\dim U^0 = \dim V - \dim U$$

Remark

Notice that $\dim U^0 = \dim V - \dim U = \dim V/U$

Proof. Start with basis of U

$$u_1, \dots, u_n \ \in U$$

Extend this to a basis of V

$$u_1, \dots, u_n, v_1, \dots, v_m \ \in V$$

Consider the dual basis of V'

$$\varphi_1, \dots, \varphi_n, \psi_1, \dots, \psi_m \ \in V'$$

We claim that

$$U^0 = \operatorname{span}\{\psi_1, \dots, \psi_m\}$$

So that

$$\begin{aligned} \dim U^0 &= m \\ &= m + n - n \\ &= \dim V - \dim U \end{aligned}$$

Then we need to prove the claim that $\psi_1, \dots, \psi_m$ is a basis of U^0.

NB $\psi_1, \dots, \psi_m$ are linearly independent by definition

$$\forall u \in U \quad u = a_1 u_1 + \cdots + a_n u_n$$

Apply ψ_i to u gives

$$\psi_i(u) = a_1\psi_i(u_1) + \cdots + a_n\psi_i(u_n) = 0 \implies \psi_i \in U^0$$

For any $\varphi \in U^0 \subseteq V'$

$$\varphi = a_1\varphi_1 + \cdots + a_n\varphi_n + \underbrace{b_1\psi_1 + \cdots + b_m\psi_m}_{\in U^0}$$

So

$$0 = \varphi(u_i) = a_i \quad \forall i = 1, \dots, n$$

And hence

$$\begin{aligned} \varphi = b_1\psi_1 + \cdots + b_m\psi_m &\implies \psi_1, \dots, \psi_m \text{ span } U^0 \\ &\implies \psi_1, \dots, \psi_m \text{ is a basis of } U^0 \end{aligned}$$

□

Corollary 12.8

For a subspace U of a finite-dimensional vector space V, we have the following facts

$$\begin{aligned} U^0 = \{0\} &\iff U = V \\ U^0 = V' &\iff U = \{0\} \end{aligned}$$

Theorem 12.9

For finite-dimensional vector spaces V and W

$$V \xrightarrow{T} W \quad W' \xrightarrow{T'} V'$$

1. $\ker T' = (\operatorname{im} T)^0$
2. $\operatorname{im} T' = (\ker T)^0$
3. $\dim \operatorname{im} T' = \dim \operatorname{im} T$
4. $\dim \operatorname{im} T' = \dim \ker T + \dim W - \dim V$